Practical - 1

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Programm - 1

Code:

```
rohitsai@MSI: ~

  GNU nano 8.7.1
#include <stdio.h>

int main() {
    printf("Welcome to operating systems\n");

    return 0;
}
```

Output:

```
rohitsai@MSI:~$ nano welcome.c
rohitsai@MSI:~$ gcc welcome.c -o welcome
rohitsai@MSI:~$ ./welcome
Welcome to operating systems
rohitsai@MSI:~$
```

Programm - 2

Code:

```
  GNU nano 8.7.1                                          hello.c
#include<stdio.h>
int main(){
char name[30];
printf("Enter your name");
scanf("%s",name);

printf("Welcome %s\n",name);
return 0;
}
```

Output:

```
rohitsai@MSI: ~
rohitsai@MSI:~$ nano hello.c
rohitsai@MSI:~$ gcc hello.c -o hello
rohitsai@MSI:~$ ./hello
Enter your name RohitSai
Welcome RohitSai
```

Programm - 3

Code:

```c
rohitsai@MSI: ~
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Process ID = %d\n", getpid());

    return 0;
}
```

Output:

```
rohitsai@MSI:~$ nano ProcessId.c
rohitsai@MSI:~$ gcc ProcessId.c -o ProcessId
rohitsai@MSI:~$ ./ProcessId
Process ID = 724
rohitsai@MSI:~$
```

Programm - 4

Code:

```
rohitsai@MSI: ~

  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main() {
    fork();

    printf("Hello\n");

    return 0;
}
```

Output:

```
rohitsai@MSI:~$ nano fork.c
rohitsai@MSI:~$ gcc fork.c -o fork
rohitsai@MSI:~$ ./fork
Hello
Hello
rohitsai@MSI:~$
```

Programm - 5

Code:

```
rohitsai@MSI: ~

  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid == 0) {
        printf("Iam child process\n");
    } else {
        printf("Iam parent process\n");
    }

    return 0;
}
```

Output:

```
rohitsai@MSI:~$ nano Parent.c
rohitsai@MSI:~$ gcc Parent.c -o Parent
rohitsai@MSI:~$ ./Parent
Iam parent process
Iam child process
rohitsai@MSI:~$
```

Programm - 6

Code:

```
rohitsai@MSI: ~
 GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid == 0) {
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    } else {
        printf("Parent PID = %d\n", getpid());
    }

    return 0;
}
```

Output:

```
rohitsai@MSI:~$ nano ParentChild.c
rohitsai@MSI:~$ gcc ParentChild.c -o ParentChild
rohitsai@MSI:~$ ./ParentChild
Parent PID = 924
Child PID = 925
Parent PID = 924
rohitsai@MSI:~$
```

Programm - 7

Code:

```
rohitsai@MSI: ~

  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid == 0) {
        printf("Child Running\n");
    } else {
        wait(NULL);
        printf("Parent Running after Child\n");
    }

    return 0;
}
```

Output:

```
rohitsai@MSI:~$ nano running.c
rohitsai@MSI:~$ gcc running.c -o running
rohitsai@MSI:~$ ./running
Child Running
Parent Running after Child
rohitsai@MSI:~$
```